TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

Reminders

Distribute first: multiply the outside number by every term inside.

Get x alone: undo + and − first, then × and ÷. Do the same thing to both sides.

Check: put your answer back in. Both sides equal → correct.

How many solutions? Same x-term and same number: infinite. Same x-term, different number: none. Otherwise: one.

1. Marcus has $35 and saves $15 per week. Which equation gives the amount he has saved, y, after x weeks?

Tip: [[y = mx + b]]: [[m]] = rate, [[b]] = start

A. $y = 35x + 15$

B. $y = -15x + 35$

C. $\mathbf{y = 15x + 35}$

Start at $35 ($b$) and add $15 each week ($m$): $y = 15x + 35$. Like test Q1.

2. What value of x makes $3(x - 2) = 12$ true?

Tip: distribute, then undo + and ×

F. 6

G. 4⅔

H. 2

Distribute: $3x - 6 = 12$, so $3x = 18$ and $x = 6$. 4⅔ forgets to multiply the 2.

3. Is $x = 2$ a solution of $5x - 1 = 9$?

> **Tip:** put the number in for [[x]]

A. No, because $10 \neq 9$

B. **Yes, because $9 = 9$**

C. Yes, because $x = 9$

Substitute: $5(2) - 1 = 10 - 1 = 9$. A true statement means 2 is a solution.

4. Which expression is equivalent to $4(x + 2) - x$?

> **Tip:** distribute, then combine like terms

F. $3x + 2$

G. $5x + 8$

H. **$3x + 8$**

Distribute: $4x + 8 - x = 3x + 8$.